

# Grammar-Derived Operation Abstraction for Cross-Project Vulnerability Detection

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## CRediT authorship contribution statement

**Quang-Vinh Dang:** Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing — original draft, Writing — review & editing, Visualization, Project administration.

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## Data availability

All code, the annotated gold set, and the scripts that regenerate every table and figure are publicly available at <https://github.com/vinhqdang/vulmorph-fed>. The study uses only publicly available corpora (BigVul, DiverseVul, PrimeVul, Devign), accessed through the public mirrors identified in the manuscript.

## Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the author used generative AI-assisted tools in order to improve the language and readability of the manuscript. After using these tools, the author reviewed and edited the content as needed and takes full responsibility for the content of the published article.

## Declaration of competing interest

The author declares that he has no known competing financial interests or personal relationships

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that could have appeared to influence the work reported in this paper.